

Padel vs Tennis on Bluesky

Social Media Network and Content Analysis

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Abstract

This report analyses the online discussion around *padel* and *tennis* on Bluesky. The objective is to understand whether the two sports form separate online communities or a partially overlapping racket-sport ecosystem. The analysis combines a user mention network with content-based evidence from sentiment analysis, emotion analysis, named entity recognition, word clouds, and hashtag frequency. The final dataset contains 1,231 posts written by 453 unique authors: 300 padel posts and 931 tennis posts. The mention network is sparse, with 471 nodes and 21 directed weighted edges. Louvain community detection identifies 451 communities with modularity equal to 0.8246, but the very small largest component indicates that the apparent separation is mainly driven by low interaction density rather than by two large cohesive blocs. Content analysis shows that tennis has more activity and more positive posts in absolute terms, while padel is represented by a smaller, more fragmented discussion.

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1 Objective

The project belongs to the social media analysis track. The goal is to analyse a topic of interest by collecting social media data, performing network analysis and content analysis, and discussing the results through visualizations.

The specific phenomenon studied in this report is the relationship between the online communities discussing *padel* and *tennis* on Bluesky. Both sports belong to the broader racket-sport domain, but they may attract different communities, accounts, hashtags, and discussion styles. The analysis is organized around four research questions:

- RQ1.** Do padel and tennis users form two separated communities, or do they overlap?
- RQ2.** Who are the most central users in each sport-related discussion?
- RQ3.** Are there bridge users that connect the two communities?
- RQ4.** Do the textual contents of the posts support the same interpretation suggested by the network structure?

The expected contribution of the project is not only descriptive. The report aims to connect structural evidence, such as centrality and community detection, with content evidence, such as sentiment, named entities, and hashtags. This is important because a sparse social graph alone may not fully represent the relationship between the two discussions.

2 Data

2.1 Collection strategy

Data were collected from Bluesky using the AT Protocol API. The collection strategy was topic-based: posts were searched separately for padel and tennis, and then merged into a single dataset. This allows a direct comparison between the two sports and makes it possible to identify users that appear in both topic datasets.

The query set was designed to include both generic keywords and topic-specific hashtags:

- **Padel queries:** `padel`, `#padel`, `#PremierPadel`, `#WorldPadelTour`.
- **Tennis queries:** `tennis`, `#tennis`, `#ATP`, `#WTA`, `#Wimbledon`, `#RolandGarros`, `#USOpen`, `#AustralianOpen`.

For each post, the following information was extracted: post URI, author handle, author DID, post text, creation date, reply count, repost count, like count, quote count, hashtags, mentions, links, topic label, and query used. Engagement was computed as:

$$\text{engagement} = \text{reply count} + \text{repost count} + \text{like count} + \text{quote count}.$$

2.2 Cleaning and enrichment

The cleaning phase normalized list-like fields, parsed dates, removed duplicated posts, and converted engagement variables into numeric values. Users were then assigned to one of three topic categories:

- **padel_only:** the user appears only in padel posts;
- **tennis_only:** the user appears only in tennis posts;
- **both:** the user appears in both padel and tennis posts.

This user-level label is essential for evaluating whether the communities overlap.

2.3 Dataset description

Table 1 reports the main post-level statistics. Tennis clearly dominates the dataset in terms of volume: it accounts for 931 posts and 365 unique authors, while padel accounts for 300 posts and 98 unique authors.

Table 1: Post-level summary by topic.

Topic	Posts	Unique authors	Avg. likes	Avg. engagement
Padel	300	98	0.767	0.930
Tennis	931	365	1.392	1.808

Figure 1 visualizes the same imbalance. This difference must be considered when interpreting the results: tennis is a larger and more active topic on Bluesky in the collected data.

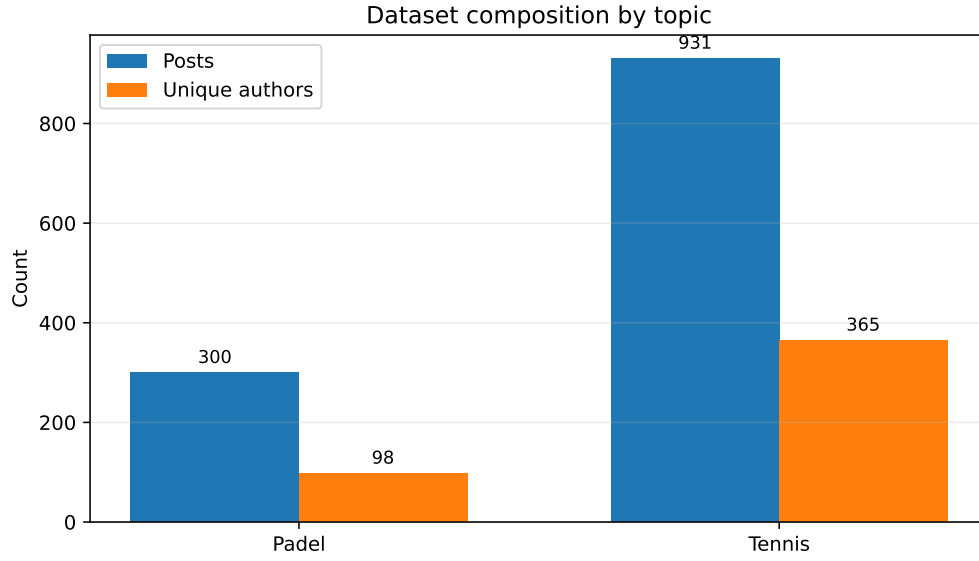


Figure 1: Dataset composition by topic.

At the user level, 88 authors are padel-only, 355 are tennis-only, and 10 appear in both topic datasets. Therefore, only 2.21% of users active in at least one of the two topics appear in both groups. This suggests limited audience overlap.

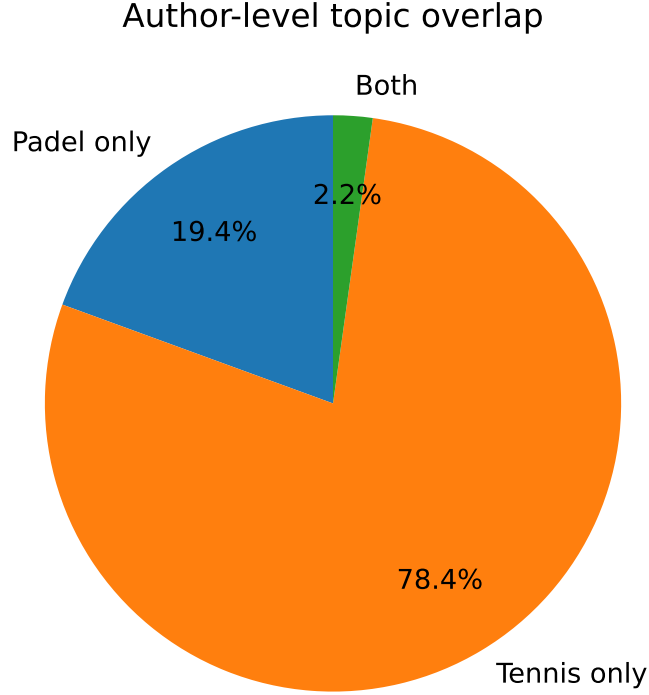


Figure 2: Author-level overlap between the padel and tennis datasets.

3 Models and Methods

3.1 Mention network construction

The main graph is a directed user mention network:

- nodes represent Bluesky users;
- a directed edge $A \rightarrow B$ indicates that user A mentioned user B in a post;
- edge weight represents the number of mentions from A to B .

This choice is appropriate because mentions are explicit interaction signals. They are more defensible than simple co-occurrence in a topic because they represent a direct relation between accounts. However, mention networks can be sparse when users mostly post without tagging others. For this reason, the network results are interpreted together with content evidence.

3.2 Graph-level metrics

For the full network and for topic-specific subnetworks, the following metrics were computed:

- number of nodes;
- number of directed edges;
- density;
- average degree;
- average clustering coefficient;

- number of weakly connected components;
- size of the largest weakly connected component.

Density is especially important here because it indicates whether the discussion behaves like a cohesive social network or like a set of isolated posts. For a directed graph with n nodes and m edges, density is:

$$D = \frac{m}{n(n-1)}.$$

3.3 Centrality analysis

Central users were identified through four complementary centrality measures:

- **In-degree centrality:** identifies users receiving many mentions, interpreted as attention or visibility.
- **Out-degree centrality:** identifies users mentioning many others, interpreted as interaction activity.
- **Betweenness centrality:** identifies users lying on many shortest paths, interpreted as potential brokers.
- **PageRank:** identifies users connected to other important users, interpreted as structural influence.

No single centrality measure is sufficient by itself. For example, PageRank can highlight accounts that receive attention from important users, while betweenness is more suitable for bridge detection.

3.4 Community detection

Community detection was performed on the undirected version of the mention network using Louvain modularity optimization. The direction of edges was ignored because the goal was to detect groups of densely connected users, regardless of who mentioned whom.

Louvain is suitable for this project because it is scalable and directly optimizes modularity. Modularity measures whether the detected groups have more internal connections than expected under a random null model. Higher values suggest stronger community separation:

$$Q = \frac{1}{2m} \sum_{ij} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j).$$

3.5 Bridge-user detection

Bridge users were defined operationally as users connecting the padel and tennis discussions. Three criteria were considered:

1. the user appears in both topic datasets;
2. the user has high betweenness centrality;
3. the user has connections to users from both topic groups.

A simple bridge score was computed as:

$$\text{bridge score} = (1 + \text{padel connections})(1 + \text{tennis connections})(1 + \text{betweenness centrality}).$$

This score is not intended as a universal metric. It is a project-specific indicator that combines topic exposure and structural brokerage.

3.6 Content analysis

Content analysis was added to complement the network evidence. The following models and techniques were used:

- **Text preprocessing:** tokenization with NLTK’s `TweetTokenizer`, removal of punctuation, URLs, stopwords, and topic keywords.
- **Sentiment analysis:** AFINN lexicon-based scoring, with posts labelled as positive, neutral, or negative.
- **Emotion analysis:** NRC Emotion Lexicon, using eight emotions: anger, anticipation, disgust, fear, joy, sadness, surprise, and trust.
- **Named Entity Recognition:** spaCy `en_core_web_sm`, extracting entities of type PERSON, ORG, GPE, EVENT, and LOC.
- **Hashtag analysis:** frequency counting and hashtag co-occurrence network.
- **Word clouds:** visual summary of the most frequent cleaned tokens.

The content models are mostly lexicon-based and rule-based. This makes them transparent and easy to interpret, but also introduces limitations: sarcasm, multilingual posts, and sport-specific slang may not be fully captured.

4 Results

4.1 Graph-level comparison

The full mention network contains 471 nodes and 21 directed weighted edges. Table 2 reports the main graph statistics.

Table 2: Graph-level statistics for the full network and topic subnetworks.

Network	Nodes	Edges	Density	Avg. degree	Components	Largest comp.
Full	471	21	0.0001	0.0892	450	5
Padel	98	0	0.0000	0.0000	98	1
Tennis	365	2	0.0000	0.0110	363	2

The most important result is the extreme sparsity of the mention network. The full graph has many weakly connected components and the largest component contains only five users. The padel subnetwork has no mention edges, meaning that padel posts in the collected dataset mostly do not mention other users. The tennis subnetwork has only two mention edges.

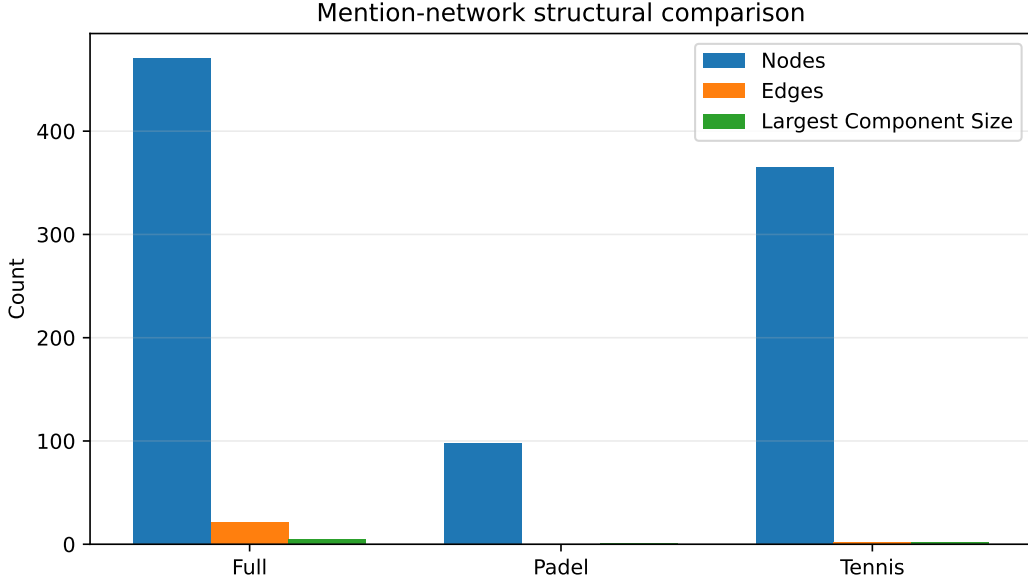


Figure 3: Structural comparison of the full, padel, and tennis mention networks.

This result suggests that Bluesky discussions on these topics are not strongly interaction-based. Many users post about padel or tennis, but they rarely mention one another. Therefore, the network should be interpreted as a sparse interaction trace rather than as a complete map of the sports communities.

4.2 Central users

Because the graph is sparse, centrality scores are low and many users have tied values. In the full network, the highest PageRank user is a mentioned-only account identified by its DID, followed by `tennistourtalk.bsky.social`. In the tennis subnetwork, `tennistourtalk.bsky.social` and `atptour.stadio.top` emerge among the most central users by PageRank. In the padel subnetwork, all top PageRank values are tied because there are no mention edges.

This produces an important methodological conclusion: in the collected data, PageRank and degree-based centrality are informative mainly for the full and tennis networks, while they are not meaningful for the padel mention subnetwork. For padel, content features such as hashtags, entities, and engagement are more informative than mention centrality.

4.3 Community detection

Louvain community detection detected 451 communities with modularity equal to 0.8246. At first sight, this modularity value is high and could suggest strong separation. However, the largest detected community contains only five users. This means that the high modularity is mainly a consequence of graph fragmentation: many isolated nodes and very small components are treated as separate communities.

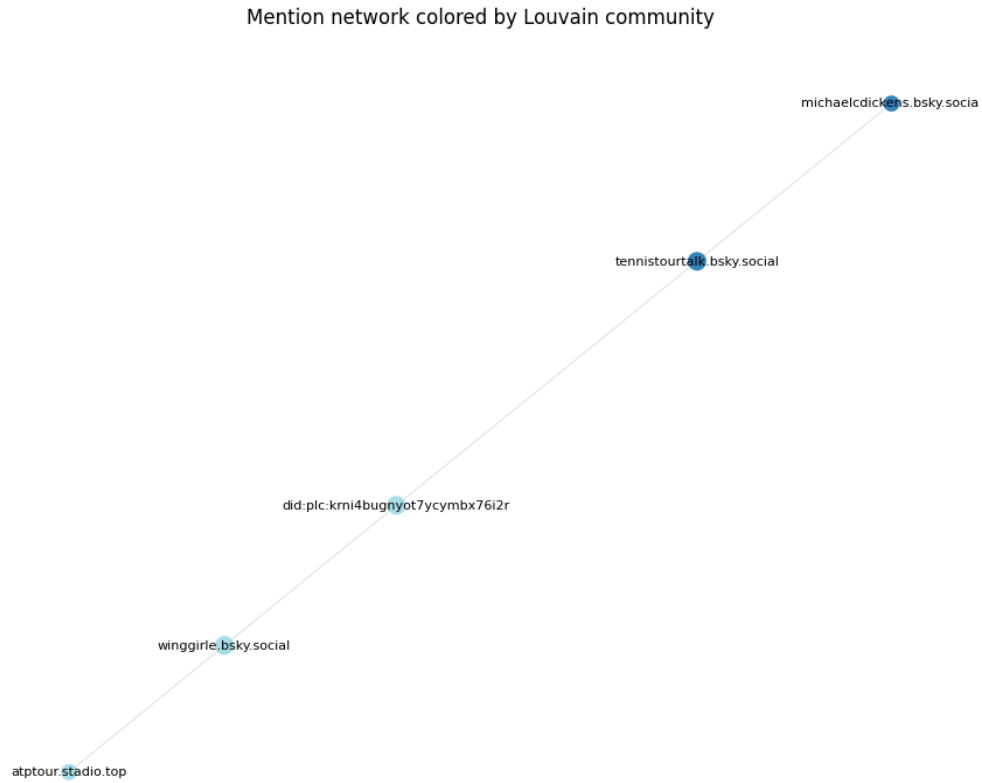


Figure 4: Full mention network colored by Louvain community.

The correct interpretation is therefore cautious. The analysis does not show two large and cohesive communities, one for padel and one for tennis. Instead, it shows a fragmented social graph with very little direct interaction. The small overlap between padel and tennis authors suggests separation at the audience level, but the network is too sparse to claim strong community polarization.

4.4 Network visualization by topic

Figure 5 shows the mention network colored by user topic type. The visualization confirms the quantitative results: the network is dominated by isolated nodes and very small components. The number of users appearing in both topics is limited.

Padel vs Tennis mention network colored by topic type

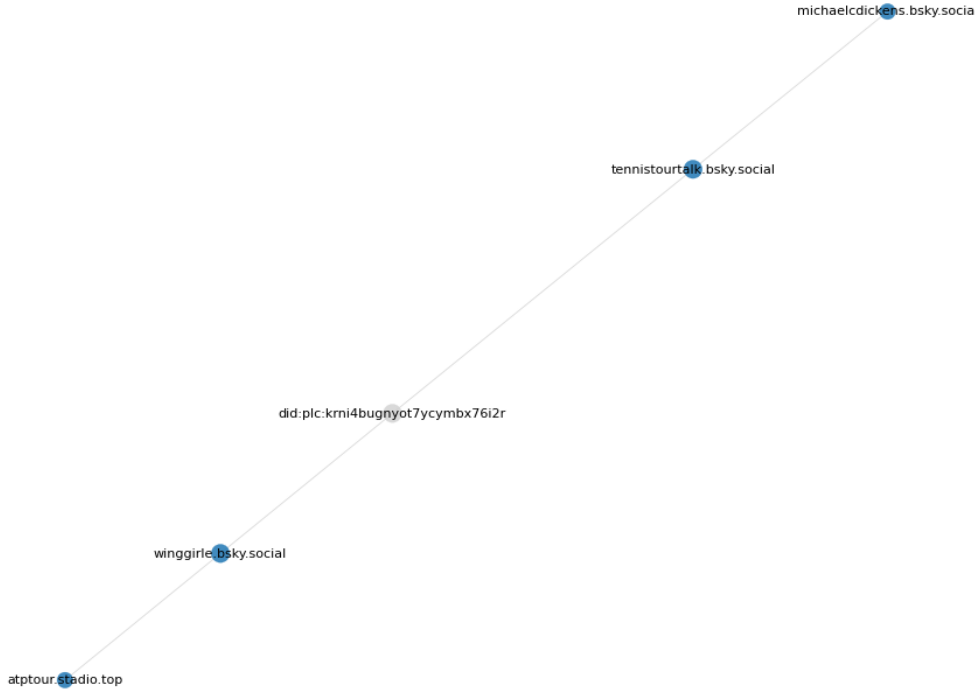


Figure 5: Full mention network colored by user topic type.

4.5 Bridge users

The bridge analysis identified only weak bridge candidates. The top bridge candidate is a mentioned-only account with bridge score 3.0000. `tennistourtalk.bsky.social` follows with bridge score approximately 2.0000 and a very low betweenness centrality value. Some users appear in both topic datasets, but they do not necessarily occupy brokerage positions in the mention graph.

This result suggests that the two topics overlap more through shared posting behaviour than through explicit mention-based interaction. In other words, there are a few users who talk about both sports, but there is no strong evidence of a stable group of structural brokers connecting the two communities.

4.6 Sentiment analysis

Table 3 reports the sentiment distribution. Tennis has more positive posts in absolute terms, but this is partly due to its larger volume.

Table 3: Sentiment distribution by topic.

Topic	Positive	Neutral	Negative
Padel	54	227	19
Tennis	373	430	128

Figure 6 shows the relative distribution. Padel is mostly neutral, while tennis has a larger share of both positive and negative posts. This is plausible because tennis includes major tournaments, elite players, match results, and emotionally charged fan discussion.

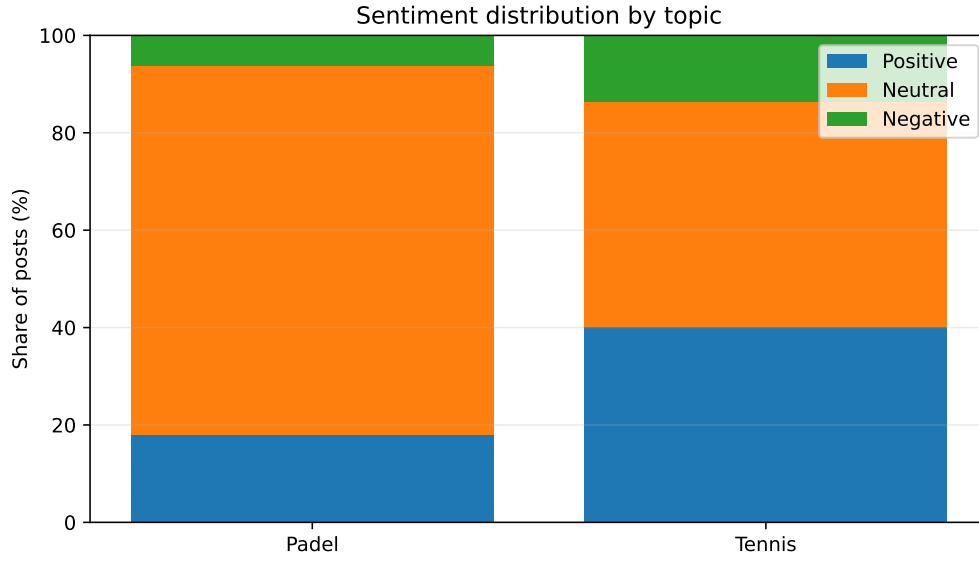


Figure 6: Relative sentiment distribution by topic.

4.7 Emotion analysis

The NRC emotion analysis provides a more detailed profile than simple polarity. Figure 7 compares the average emotion frequencies across the two topics. This visualization should be interpreted as a lexical signal: it shows which emotion-associated words appear more frequently, not the true psychological state of users.

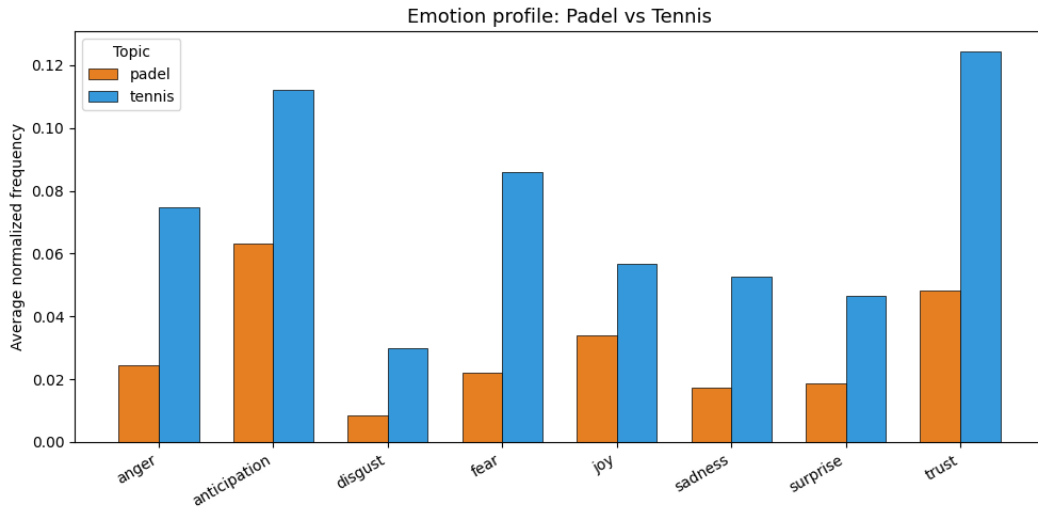


Figure 7: Emotion profile of padel and tennis posts using the NRC Emotion Lexicon.

4.8 Named Entity Recognition

Named Entity Recognition was used to identify the people, organizations, places, events, and locations most frequently mentioned in each topic. Figure 8 shows the top entities for padel and tennis.

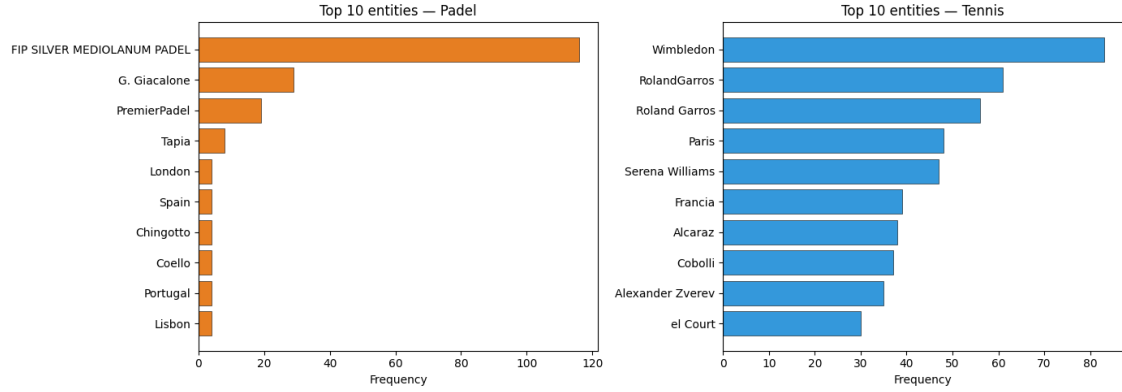


Figure 8: Top named entities in padel and tennis posts.

The entity analysis is useful because it moves beyond the words *padel* and *tennis*. Tennis discussion is expected to contain more references to tournaments, tours, and professional players, while padel discussion may include tournament circuits, clubs, and lifestyle-related references.

4.9 Hashtag frequency

Hashtags are explicit labels chosen by users and therefore provide a strong signal of topic framing. Table 4 reports the top hashtags by topic.

Table 4: Top hashtags by topic.

(a) Padel		(b) Tennis	
Hashtag	Count	Hashtag	Count
padel	152	tennis	276
premierpadel	53	rolandgarros	219
padelfun	39	atp	201
foryou	38	wta	177
fyp	38	wimbledon	131
padeladdict	38	australianopen	130
padeltime	38	usopen	125
padelvideos	38	grandslam	64
voorjou	38	frenchopen	55
padelmania	37	paris	44

The tennis hashtags are strongly connected to professional circuits and Grand Slam events. The padel hashtags include **premierpadel**, but also more social-media-oriented tags such as **foryou**, **fyp**, and **padelvideos**. This suggests that tennis discussion is more event-driven, whereas padel discussion in the collected data contains more promotional or lifestyle-style content.

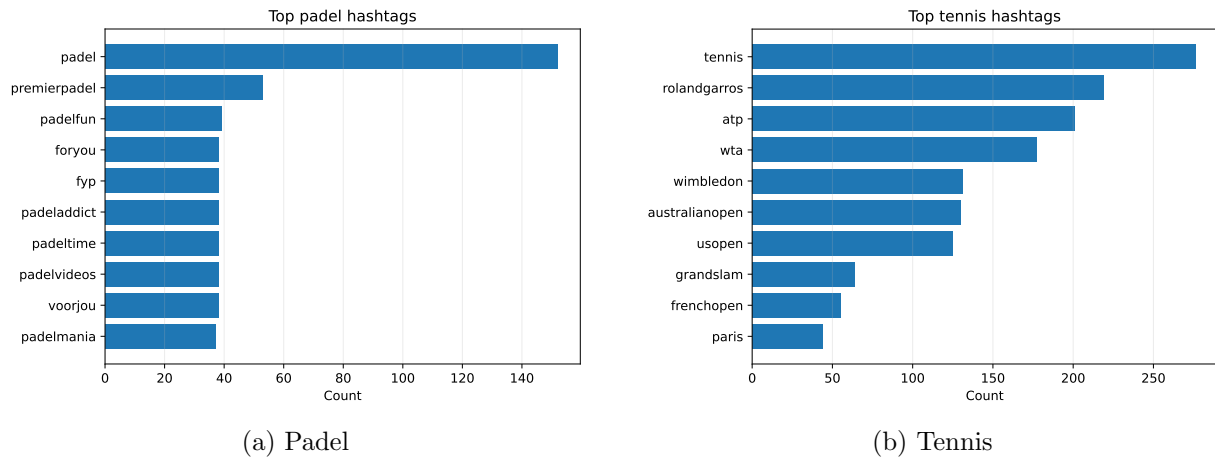


Figure 9: Top hashtag frequencies by topic.

4.10 Word clouds and hashtag co-occurrence

Word clouds summarize the most frequent cleaned tokens. They are not statistical models, but they are useful for presentation because they show the vocabulary of each topic at a glance.

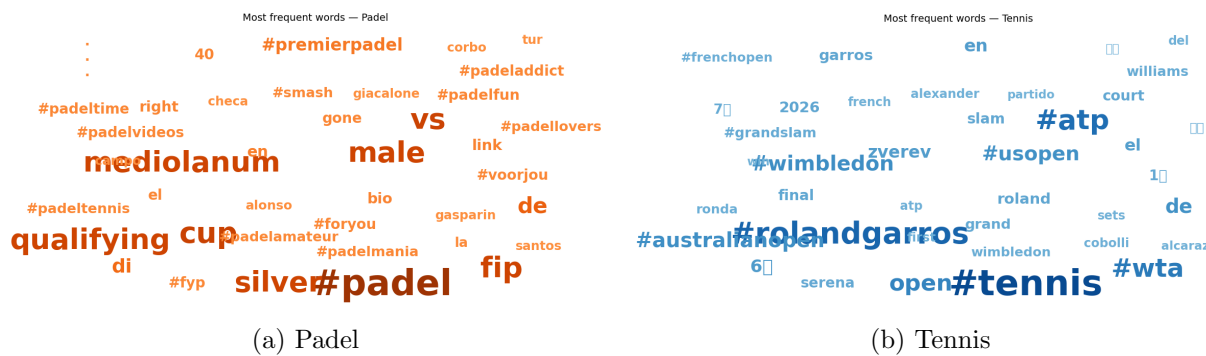


Figure 10: Word clouds after text preprocessing.

A hashtag co-occurrence network was also built. In this graph, hashtags are nodes and two hashtags are connected if they appear in the same post. This helps identify shared vocabulary and clusters of hashtags around events, tours, or social-media trends.

5 Discussion

RQ1: Do padel and tennis communities overlap? The author-level overlap is limited: only 10 users appear in both topic datasets, corresponding to 2.21% of users active in at least one topic. This supports the idea that padel and tennis discussions are mostly separated at the audience level. However, the mention network is too sparse to describe two cohesive communities.

RQ3: Are there bridge users? Bridge evidence is weak. Some users appear in both topic datasets, but they do not form a strong brokerage structure in the mention network. The highest bridge scores are low and are partly driven by mentioned-only nodes. Therefore, bridge users exist only in a limited operational sense.

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6 Limitations

The project has several limitations:

- **Sampling bias:** the dataset depends on the selected queries and on the Bluesky search API.
- **Topic imbalance:** tennis has many more posts than padel, which affects comparisons.
- **Sparse mentions:** many users discuss the topics without mentioning others, making the mention network fragmented.
- **Lexicon-based sentiment:** AFINN and NRC may not capture sarcasm, multilingual expressions, or sport-specific vocabulary.
- **NER noise:** spaCy may misclassify short posts, usernames, scores, or hashtags as named entities.

These limitations do not invalidate the analysis, but they require cautious interpretation. In particular, the absence of mention edges should not be interpreted as absence of community. It only means that explicit mention-based interaction is limited in the collected dataset.

7 Conclusion

The analysis suggests that padel and tennis on Bluesky are mostly separated at the audience and content levels, but not because they form two large and polarized interaction communities. Instead, the mention network is extremely sparse and fragmented. Tennis appears as the larger and more event-driven discussion, while padel appears smaller and more weakly connected in the collected data.

The strongest evidence of separation comes from the low author overlap and from the different hashtag ecosystems. The weakest evidence comes from community detection, because Louvain modularity is high mainly due to graph fragmentation. Overall, the project shows why network analysis and content analysis should be combined: the network alone reveals limited direct interaction, while the content analysis explains what each topic is actually about.